

# Sean Luka Girgis

Production Engineer / SRE | Python, Oracle/SQL, Monitoring, RCA & Data Pipeline Reliability

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## Professional Summary

Senior production engineering, SRE-adjacent, data reliability, and performance professional with 20+ years of enterprise technology experience across Python, SQL, Oracle, PL/SQL, production support, monitoring, incident response, root cause analysis, capacity planning, automation, data pipeline reliability, CI/CD concepts, and cloud data workflows. Strong background supporting regulated enterprise environments, troubleshooting production data issues, optimizing SQL/reporting layers, documenting runbooks, improving monitoring visibility, and translating operational signals into practical remediation. Best aligned to Production Engineer / SRE roles requiring Python/SQL/Oracle, monitoring, RCA, capacity planning, automation, data pipeline support, CI/CD collaboration, and operational reliability; Snowflake, Java/Spring Boot microservices, SAP FI/CO, ABAP, SAP Solution Manager, and deep Azure DevOps/Jenkins ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

## Professional Experience

### Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported production monitoring, capacity, and data reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and large-scale infrastructure telemetry sources.
- Built Python/Pandas/PySpark and SQL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw performance signals into validated dashboards, reports, forecasting datasets, and executive insights.
- Performed production troubleshooting and RCA-style analysis by tracing source feeds, transformation logic, SQL behavior, metric behavior, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Developed validation and monitoring checks for row counts, nulls, unmatched joins, abnormal trend changes, freshness, and output consistency to improve pipeline reliability and reduce silent data issues.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Applied capacity planning and forecasting workflows using Prophet, scikit-learn, Python, SQL, and telemetry datasets to identify bottleneck risk and support proactive infrastructure planning.
- Created runbooks, metric definitions, validation checkpoints, support procedures, troubleshooting notes, and operational documentation to improve repeatability and incident response.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize issues, explain findings, coordinate fixes, and improve operational reliability.

### Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed performance analysis and monitoring for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.

- Correlated load-test behavior with APM telemetry to identify bottlenecks, user-impacting symptoms, and improvement areas before production impact.
- Led the FAST project, mining real-user and synthetic monitoring metrics to identify performance improvement opportunities adopted by engineering teams.
- Created dashboards and reporting views that converted monitoring and test signals into actionable performance, reliability, and stakeholder insights.

#### **SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)**

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts used by application, middleware, infrastructure, and operations teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable operational processes.
- Provided mentoring, onboarding support, troubleshooting guidance, and documentation to improve team adoption of monitoring and performance-analysis practices.

#### **Performance Test Engineer at AT&T (2010-08 – 2011-07)**

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, NFR-style evaluation, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

#### **Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)**

- Led validation and performance-focused cutover support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OC CI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported test planning, phased validation, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

#### **Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)**

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

#### **Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)**

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro\*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

## Education

### **Post-Graduate Diploma in Computer Engineering Technology / Computer Science**

Humber College, Toronto, Canada

### **Bachelor of Science in Civil Engineering**

Zagazig University, Egypt

## Skills

### Flagship Projects

#### **Production Data Quality and RCA Workflow**

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and troubleshooting documentation.

**Technologies:** Python, SQL, Pandas, Oracle, Data Validation, RCA, Runbooks

#### **HorizonScale — Capacity Forecasting and Monitoring**

Built Python/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights for proactive risk mitigation.

**Technologies:** Python, Pandas, PySpark, SQL, Prophet, BMC TrueSight, Data Validation

#### **Serverless Lakehouse Platform (AWS)**

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

**Technologies:** AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality

#### **Enterprise APM Reporting Automation**

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

**Technologies:** CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation